

Analysis of Variance (ANOVA)

Types of ANOVA (3 Types)

① One way ANOVA: One factor with at least 2 levels, these levels are independent.

Eg:- Doctor wants to test a new medication to decrease headache.

They split the participants in 3 conditions [10 mg, 20 mg, 30 mg]

Doctors ask participants to rate the headache [1 - 10]

Medication → Factor

10 mg	20 mg	30 mg
5	7	2
3	4	6
-	-	-
-	-	-

② Repeated Measure Anova :-

One factor with atleast 2 levels, levels are dependent.

Running $\rightarrow$ Factor

levels $\rightarrow$	Day 1	Day 2	Day 3
8		5	4
7		4	9
-		-	-

③ Factorial ANOVA: Two or more factor (each of which with atleast) 2 levels, levels can be independent and dependent.

Running $\rightarrow$ Factor

levels $\downarrow$	Day 1	Day 2	Day 3
male	8	5	4
9		4	3
2		4	6
Female	7	8	3